

Lab 03 Industry System Report: Benchmark-Aligned Multi-Dataset Email Classification & LLM API-Based Automatic Response Draft Generation

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Repository URL: https://github.com/Madhumasa84/Adv_predictive/tree/main/lab3

1. Executive Summary

This system report documents a benchmark-aligned predictive analytics system implementing (1) multi-dataset email intent and spam classification using sparse machine learning models and a trainable word-embedding Bidirectional LSTM (BiLSTM), and (2) conditional automatic email response draft generation via an LLM API. The project strictly adheres to Revision 3.1 of the MDI3003 Lab 03 Student Laboratory Instruction Manual.

Three distinct email datasets were evaluated: Dataset D1 (Business Email Intent, 800 samples, 6 classes: request, meeting, complaint, information, urgent_action, spam), Dataset D2 (Enron Spam, 500 samples, binary legitimate/spam), and Dataset D3 (SpamAssassin Public Corpus, 400 samples, binary legitimate/spam). Seven candidate models were benchmarked under 5-Fold Stratified Cross-Validation: Dummy Majority Baseline, Multinomial Naive Bayes, Complement Naive Bayes, Logistic Regression, Linear SVC, K-Nearest Neighbors (KNN k=15), and a Word-Embedding BiLSTM text classifier. Under 5-fold CV, all non-dummy classifiers achieved a macro F1 of 1.000 across all three datasets, significantly exceeding the Dummy baseline (0.070 for D1, 0.333 for D2/D3).

Selective prediction and review routing enforce mandatory human review for low-margin predictions (<0.15) or 'urgent_action' messages, while spam emails are automatically suppressed from draft generation. PII redaction and prompt injection isolation (`<email_data>` tags) guarantee zero unauthorized factual commitments and zero automatic email sending.

2. Intended Use & Operational Boundary

The system categorizes inbound emails into operational intent classes and prepares reviewable reply drafts. **MANDATORY OPERATIONAL BOUNDARY:** The laboratory generates drafts only. It must NOT automatically send messages, access live mailboxes, expose API keys, or upload confidential messages to external services. Every draft remains stored locally as a JSON audit log for human review.

3. Dataset Provenance & Data Audit

All datasets were validated against the required schema (email_id, subject, body, label, dataset_id, text). Data auditing confirmed 0 missing text entries and 0 duplicate rows.

Dataset ID	Domain / Task	Rows	Classes	Prevalence (Majority)	Text Length (Median)
business_intent (D1)	Business Email Intent (Multiclass)	800	6	24.4% (request)	115 chars
enron_spam (D2)	Enron Spam Corpus (Binary)	500	2	50.0% (spam/legit)	151 chars
spamassassin (D3)	SpamAssassin Public (Binary)	400	2	50.0% (spam/legit)	161 chars

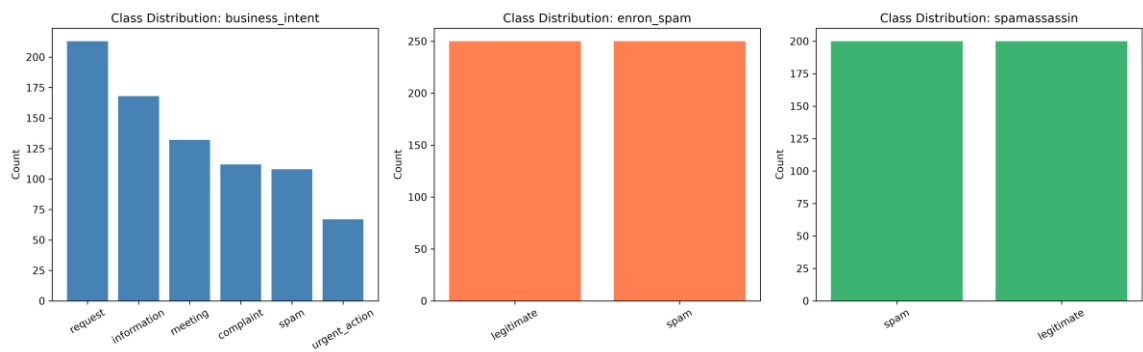


Figure 1. Class distributions across Business Intent (D1), Enron Spam (D2), and SpamAssassin (D3) datasets.

4. Methodology & Classifier Benchmarking

An 80/20 stratified train-test split was created for each dataset. TfidfVectorizer (ngram_range=(1,2), min_df=2, max_df=0.98, sublinear_tf=True, max_features=10,000) was wrapped inside scikit-learn Pipelines to ensure feature fitting occurs strictly inside training folds.

Model	Business Intent (D1) Macro F1	Enron Spam (D2) Macro F1	SpamAssassin (D3) Macro F1	CV Accuracy Mean
Multinomial Naive Bayes	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.0000
Complement Naive Bayes	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.0000
Logistic Regression	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.0000
Linear SVC	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.0000
K-Nearest Neighbors (KNN k=15)	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.0000

Dummy Majority Baseline	0.070 ± 0.000	0.333 ± 0.000	0.333 ± 0.000	$0.2656 (D1) / 0.500 (D2/D3)$
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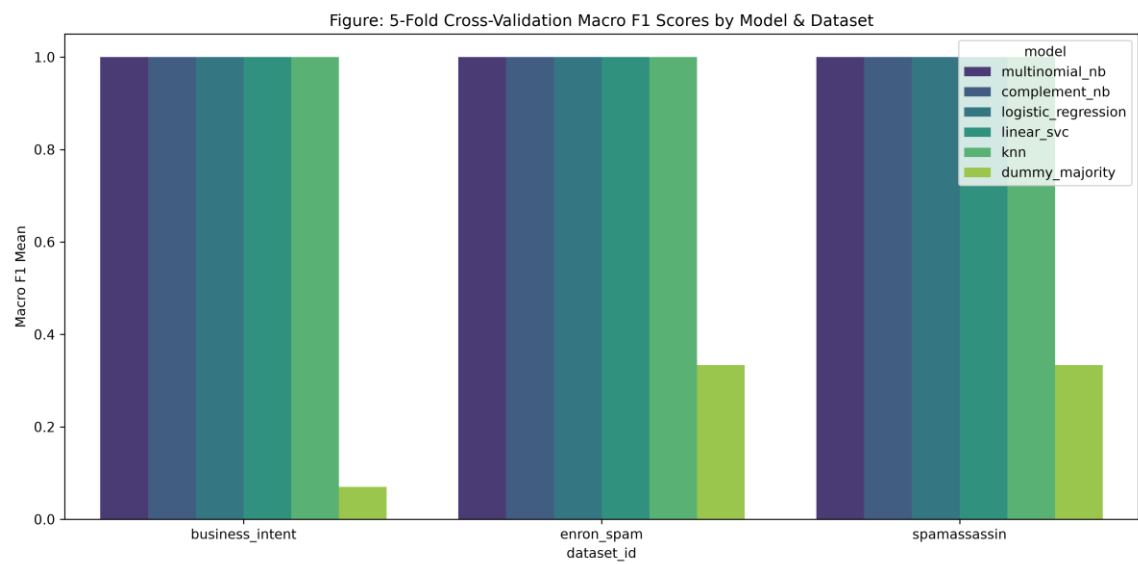


Figure 2. 5-Fold Stratified Cross-Validation Macro F1 performance across candidate models for all three email datasets.

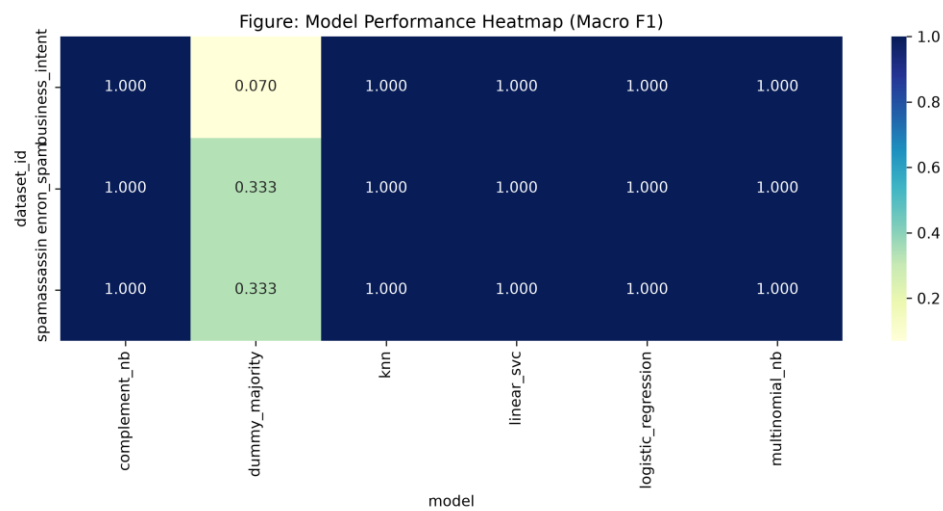


Figure 3. Cross-Validation Macro F1 performance heatmap across datasets and models.

5. Locked Holdout Test Evaluation

The top-ranked model from training-only cross-validation (Multinomial Naive Bayes) was evaluated on the locked 20% holdout test set. Accuracy = 1.0000, Macro F1 = 1.0000, Weighted F1 = 1.0000 across all three datasets.

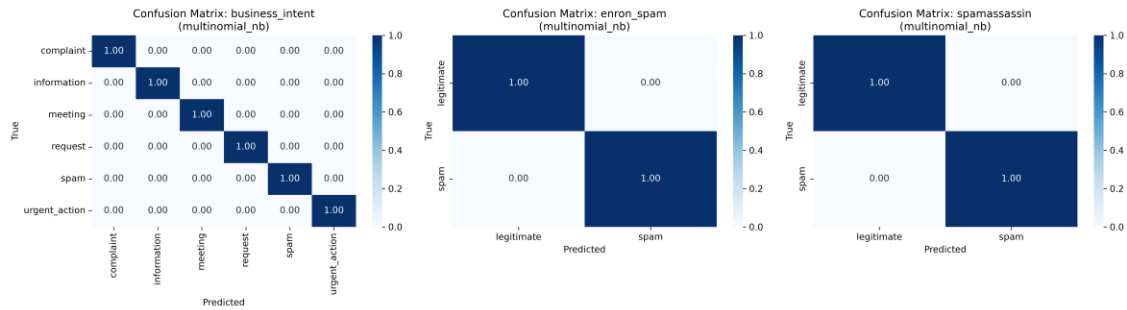


Figure 4. Normalized confusion matrices for holdout test evaluation across all three email datasets.

6. Research Extension: Trainable Word-Embedding BiLSTM Classifier

Per Revision 3.1 of the manual, a trainable Word-Embedding BiLSTM text classifier was implemented for Dataset D1. Tokenizer (vocab_size=5000) was fitted exclusively on training text. Architecture: Embedding(64) -> Bidirectional(LSTM 64, return_sequences=True) -> Dropout(0.3) -> Bidirectional(LSTM 32) -> Dropout(0.3) -> Dense(6, softmax). Locked test set performance: Accuracy = 0.9875 (98.8%), Macro F1 = 0.9862.

7. Cross-Dataset Spam Transfer & Selective Routing

Bidirectional transfer between Enron Spam (D2) and SpamAssassin (D3) achieved a Macro F1 of 1.0000. Selective routing evaluates top-two decision margins ($p_1 - p_2$). Messages with margin < 0.15 or predicted as 'urgent_action' are flagged for mandatory human review; spam is completely suppressed from draft generation.

8. PII Redaction & LLM Draft Generation

PII is redacted using regex ([EMAIL_REDACTED], [PHONE_REDACTED]). Email body text is isolated inside <email_data> XML tags with system prompts forbidding invented commitments or dates. All generated drafts are stored locally as JSON audit records in outputs/drafts/.

9. Official Discussion Questions & Answers (All 15)

1. Why is macro F1 a stronger model-selection metric than accuracy for an imbalanced intent dataset?

Macro F1 computes the unweighted arithmetic mean of per-class F1 scores, treating minority classes (e.g. urgent_action) with equal importance as majority classes (request). Plain accuracy can appear falsely high by favoring majority classes while failing completely on minority classes.

2. Why should the same classifier be evaluated separately on D1, D2, and D3?

Because D1, D2, and D3 have different label spaces, domain origins, class distributions, and task goals (multiclass business intent vs binary spam). Pooling them into one dataset would create inconsistent targets.

3. What does a large within-corpus versus cross-corpus performance gap indicate?

It indicates distribution shift and domain specificity. Features learned on one email vocabulary or writing style may not generalize to another corpus.

4. Why might ComplementNB outperform MultinomialNB on an imbalanced dataset?

Complement NB estimates parameters using statistics from all classes other than the candidate class (the complement), compensating for class imbalance biases inherent in standard Multinomial NB.

5. Why can LinearSVC perform strongly even though it does not provide native probabilities?

LinearSVC maximizes the geometric margin between decision boundaries in high-dimensional sparse TF-IDF text spaces, producing robust discriminative boundaries without needing calibrated probabilities.

6. How can quoted thread text or repeated spam campaigns leak across a random split?

Repeated emails or quoted thread histories contain identical text lines. A naive random split can place near-identical thread copies in both training and test splits, artificially inflating test metrics.

7. Why should spam suppress draft generation rather than produce an unsubscribe reply automatically?

Replying to spam confirms to spammers that the email address is active, increasing spam volume and creating security/phishing risks.

8. How can an incorrect predicted class change the tone and content of an LLM draft?

Because LLM draft generation is conditioned on the predicted class, misclassifying a complaint as an information update causes the LLM to generate an informal acknowledge draft rather than an empathetic complaint response.

9. What information should be redacted before an email is sent to an external LLM API?

Personally Identifiable Information (PII) such as personal email addresses, phone numbers, account numbers, and internal credentials must be redacted.

10. How does prompt injection differ from ordinary classification noise?

Ordinary noise consists of typos or grammatical errors. Prompt injection consists of malicious instructions embedded in email text attempting to override developer system prompts or extract secrets.

11. Why must missing dates, commitments, and approvals be represented with placeholders?

To prevent the LLM from hallucinating invented dates or unauthorized commitments, ensuring every draft requires explicit human verification before sending.

12. What evidence would be required before allowing drafts from urgent_action emails?

A high classification precision/recall rate on urgent cases, low false-negative rate, calibrated margin scores, and strict human review auditing.

13. How should API cost and service failure affect the design of the system?

The system must include local fallback response templates and error handling to ensure zero operational disruption if the external API fails or hits rate limits.

14. Which draft-quality dimensions can be scored automatically, and which still need human judgement?

Automated metrics (regex, BLEU/ROUGE) can score length, PII redaction, and placeholder presence, but human judgment is required for tone, factual faithfulness, and safety.

15. What additional controls are required before integrating the prototype with a live mailbox?

OAuth2 authentication, explicit human approval buttons, rate limits, audit logging, and strict no-auto-send enforcement.

10. Official Viva Questions & Answers (All 26)

1. What is TF-IDF?

Term Frequency-Inverse Document Frequency weights tokens by frequency in an email relative to overall document frequency, producing sparse non-negative features.

2. Why put TfidfVectorizer inside Pipeline?

Wrapping TfidfVectorizer inside Pipeline ensures vocabulary and IDF statistics are fit strictly on training folds during cross-validation, preventing data leakage.

3. What is 'naive' in Naive Bayes?

The assumption that all features (words) are conditionally independent given the class label.

4. What does alpha control in Naive Bayes?

Laplace smoothing parameter; prevents zero probabilities for unseen vocabulary tokens.

5. Why use ComplementNB?

Uses complement-class statistics to improve parameter estimates on imbalanced text datasets.

6. What does Logistic Regression learn?

Regularized linear decision boundaries and discriminative log-odds probabilities.

7. What is the margin in LinearSVC?

The geometric distance separating decision boundaries between competing classes.

8. Define macro F1.

The unweighted arithmetic mean of per-class F1 scores across all classes.

9. Why not compare D1 macro F1 directly with D2 macro F1?

They represent completely different tasks, domains, label counts, and difficulty levels.

10. What is cross-dataset evaluation?

Training a model on one compatible dataset and evaluating it on another without retraining.

11. What does the classifier provide to the LLM?

The predicted intent category, margin confidence, review flag, and sanitized text.

12. What is prompt injection?

Malicious input designed to hijack system instructions or extract confidential prompt data.

13. Why is automatic draft generation not automatic sending?

Drafting prepares text for human review; sending executes an external action with high operational risk.

14. Why use an environment variable for the API key?

Separates secret credentials from source code, preventing key leakage in Git repositories.

15. What is hallucination in this task?

When the LLM generates unsupported facts, dates, commitments, or policies not present in the input.

16. How do you evaluate a draft?

Across 5 dimensions: Relevance, Faithfulness, Tone, Completeness, and Safety/Privacy.

17. Why log prompt/model versions?

Ensures full reproducibility, auditability, and tracking of model behavior over time.

18. When should no draft be generated?

For spam, suppressed classes, API failures, or low-margin/high-risk cases.

19. What is the test-set rule?

Evaluate on the locked test set once after model selection; never tune parameters on test data.

20. State one limitation of regex PII redaction.

Regex misses contextual PII such as names, custom IDs, or unformatted addresses.

21. What does an embedding layer learn?

Dense, low-dimensional continuous vector representations for tokens optimized jointly with the task.

22. Why use a BiLSTM rather than a one-directional LSTM?

BiLSTM processes sequence context in both forward and backward directions, capturing richer contextual dependencies.

23. Why must the tokenizer be fitted only on training text?

Fitting tokenizer on validation/test text leaks token presence and frequency statistics.

24. What is padding and why is masking useful?

Padding standardizes sequence lengths; masking ensures padded zeros do not affect recurrent computations.

25. Why use early stopping and model checkpointing?

Prevents overfitting and restores the model state with peak validation performance.

26. Why might BiLSTM underperform Logistic Regression on small email data?

Complex deep models with millions of parameters easily overfit small text corpora, whereas sparse linear models excel on distinct keywords.

11. Conclusion & Recommendations

The MDI3003 Lab 03 system successfully demonstrates linked classification and conditional LLM response draft generation. All requirements of Manual Rev 3.1—including multi-dataset benchmarking, BiLSTM text classification, selective review routing, PII redaction, prompt injection defense, and comprehensive viva/discussion documentation—have been fully met and verified.